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# Expansion of low-load operating range by mixture stratification in a natural gas-diesel dual-fuel premixed charge compression ignition engine

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ABSTRACT

Dual-fuel premixed charge compression ignition (DF-PCCI) combustion can achieve low nitrogen oxides ($NO_X$) and particulate matter (PM) emissions for wide ranges of engine operations. However, the deterioration in thermal efficiency, and hydrocarbon (HC) and carbon monoxide (CO) emissions at low loads were recognized as the barriers for expanding the low-load operating range. In this study, the causes of the barriers were investigated and a mixture preparation strategy was suggested for overcoming the barriers in a natural gas (NG)-diesel DF-PCCI engine. Combustion and energy balance analysis was conducted to evaluate the strategy. Baseline DF-PCCI was determined by combinations of diesel start of injection (SOI) and NG substitution ratio (SR) at low loads from 0.3 to 0.6 MPa indicated mean effective pressure (IMEP). An increase in the homogeneity of a fuel-air mixture in the baseline DF-PCCI effectively reduced the $NO_X$ and PM emissions but increased the HC and CO emissions in each low-load operation. As the engine load was decreased, the formation of an overly-lean mixture intensified the effects of the mixture homogeneity. Therefore, the thermal efficiency, and HC and CO emissions deteriorated at 0.3 MPa IMEP. A mixture stratification strategy was established to increase the local equivalence ratio and reactivity of the fuel-air mixture. The strategy was realized by a retarded diesel SOI, a lower NG SR, and a higher exhaust gas recirculation rate. The strategy increased the degree of constant volume combustion by enhancing the combustion performance. The enhanced combustion reduced the combustion loss, and thus, improved the thermal efficiency. The HC and CO emissions also decreased mainly due to the improved combustion and the reduced mass flow rates of the NG.

## 1. Introduction

Diesel engines have been widely used as power sources in transportation because of their high thermal efficiency and torque. However, as standards for carbon dioxide ($CO_2$) and other harmful emissions are stringent in transportation, there is a demand for improved efficiency and reduced exhaust emissions in diesel engines. In particular, the reduction in nitrogen oxides ($NO_X$) and particulate matter (PM) emissions from the diesel engines is an issue of growing importance. Currently, diesel engines should be equipped with after-treatment devices to meet the EURO VI emission legislation. Although the after-treatment devices are attractive tools for reducing the $NO_X$ and PM emissions, their implementation increases vehicle cost and deteriorates the fuel economy [1]. Therefore, reducing emissions through in-cylinder combustion technology is also required to ease the burden of after-treatment systems.

For conventional diesel combustion (CDC), $NO_X$ is generated at the boundary of the diffusion flame, which is characterized by a stoichiometric fuel-air mixture and high temperature [2]. The PM is formed in regions with high equivalence ratios and high temperatures. In an attempt to reduce the $NO_X$ and PM emissions from the CDC, diesel-based in-cylinder combustion concepts, such as homogeneous charge compression ignition (HCCI), premixed charge compression ignition (PCCI), and low temperature combustion (LTC), have been developed. These

*Abbreviations:* aTDC, after top dead center; CAD, crank angle degree; CDC, conventional diesel combustion; CO, carbon monoxide; CO2, carbon dioxide; COV, coefficient of variation; DF-PCCI, dual-fuel premixed charge compression ignition; DOCV, degree of constant volume; EGR, exhaust gas recirculation; HC, hydrocarbon; HCCI, homogeneous charge compression ignition; HRR, heat release rate; IMEP, indicated mean effective pressure; ITE, indicated thermal efficiency; LHV, lower heating value; LTC, low temperature combustion; MPRR, maximum pressure rise rate; NG, natural gas; NOX, nitrogen oxides; PCCI, premixed charge compression ignition; PPC, partially premixed combustion; PM, particulate matter; SOI, start of injection; SR, substitution ratio; TDC, top dead center; THC, total hydrocarbon

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concepts reduced the $NO_X$ and PM emissions simultaneously due to the reduction in combustion temperature of premixed fuel-air charges [3]. Fuel conversion efficiency was improved in these combustion concepts mainly through the optimized thermodynamic cycles and the reduction in heat losses [4]. However, there have been some challenges in commercializing the combustion concepts. The combustion phasing control was limited because chemical kinetics dominated the auto-ignition of the premixed fuel-air mixture [5]. The operating range was confined to low loads because the rapid volumetric combustion of the premixed mixture resulted in high levels of in-cylinder pressure and maximum pressure rise rate (MPRR) during high-load operations [6]. Gasoline-based in-cylinder combustion concepts, such as gasoline HCCI and gasoline partially premixed combustion (PPC), were suggested to address the limited high-load operating range in the diesel-based combustion concept. Gasoline is beneficial for fuel-air mixing because of its high volatility and high resistance of auto-ignition [7]. Therefore, a higher engine load was achieved in the gasoline PPC than in the diesel-based in-cylinder combustion [8]. However, combustion instability, and excessive hydrocarbon (HC) and carbon monoxide (CO) emissions were drawbacks in low-load operations of the gasoline PPC [9]. The spark-assisted combustion concept [10] and negative valve overlap [11] were effective in lessening these drawbacks, but the $NO_X$ emissions reached unacceptable levels. When only one fuel was applied to the in-cylinder combustion concept, the engine operation was limited to specific ranges [12].

Dual-fuel premixed charge compression ignition (DF-PCCI) concepts blend two fuels having different reactivity in the combustion chamber [13]. In the DF-PCCI concepts, the equivalence ratio and reactivity of a fuel-air charge were controlled by modulating the fuel start-of-injection (SOI) timing and the blending ratio between the two fuels [14]. While single-fuel in-cylinder combustion concepts stratified the equivalence ratio alone to achieve the high-load operations [15], fuel reactivity, in addition to the equivalence ratio, stratification was employed for expanding the high-load operating range in the DF-PCCI concepts [16]. The reactivity stratification led to the staged auto-ignition of the high-reactivity fuel and the low-reactivity fuel, which reduced the peak heat release rate (HRR) and MPRR [17]. The reactivity and equivalence ratio stratification in the DF-PCCI concept also led to an improvement in the combustion phasing control [18]. Previous studies have shown that simultaneous reduction in $NO_X$ and PM emissions can be achieved in gasoline-diesel DF-PCCI engines at wide ranges of engine speeds and loads [19–21]. A combination of diesel SOI and a blending ratio, without exhaust gas recirculation (EGR), achieved low $NO_X$ and PM emissions at low loads [19]. An advanced diesel SOI, a higher gasoline portion, and a higher EGR rate were implemented as the load was increased in order to reduce the high in-cylinder pressure and MPRR [20]. Although the engine was operated at a wider range of engine speeds and loads in gasoline-diesel DF-PCCI concepts than in single-fuel in-cylinder combustion concepts, the full-load operation was limited [21]. Miller cycle strategies [22] and a late diesel SOI [23] effectively reduced the peak in-cylinder pressure and MPRR at the full-load operation, but the $NO_X$ levels increased. The engine operation at low loads also exposed shortcomings in the gasoline-diesel DF-PCCI concepts. The thermal efficiency and HC and CO emissions deteriorated substantially more in the DF-PCCI concept than in diesel CDC [24] and diesel LTC [25].

Natural gas (NG) is considered as an alternative to conventional petroleum because the low-carbon fuel can potentially lead to a reduction in $CO_2$ emissions in transportation [26]. The NG in the DF-PCCI combustion concept is an attractive option because its high octane number can be an advantage in expanding the high-load operating range. Walker et al. [27] used experimental methods to evaluate the potential for expanding the high-load limit in NG-diesel DF-PCCI mode. A comparative analysis between gasoline-diesel and NG-diesel DF-PCCI modes was conducted in a single-cylinder, heavy-duty engine by imposing a constraint of MPRR. A lower peak HRR and wider combustion duration led to the achievement of a higher load operation in the NG-diesel DF-PCCI mode than in the gasoline-diesel DF-PCCI mode. The combustion behavior was due to its lower reactivity than that of gasoline. Nieman et al. [28] simulated an NG-diesel DF-PCCI engine at broad ranges of engine operations using computational fluid dynamics (CFD) software with the CHEMKIN. A larger reactivity difference of NG and diesel than that of gasoline and diesel allowed the engine operation up to 23 bar indicated mean effective pressure (IMEP) with low $NO_X$ and PM emissions. However, the authors also pointed out that the low reactivity of NG can be a disadvantage at low loads. Although low $NO_X$ and PM emissions were achieved with diesel SOI and the blending ratio of NG and diesel at the load range from 4 to 13.5 bar IMEP, the combustion mode suffered from the deterioration in gross indicated efficiency, and HC and CO emissions at 4 bar IMEP. Doosje et al. [29] explored the potential for expansion to low-load operating range in the DF-PCCI concept fueled with NG and diesel. As the engine load was decreased, low $NO_X$ and PM emissions were retained. However, the fuel consumption increased sharply at the lowest load investigated mainly because of the increased losses in friction and combustion. Ansari et al. [30] used empirical models to compare NG-diesel DF-PCCI mode to CDC in a four-cylinder, light-duty engine. The models were created based on a regression analysis of experimental data to assess the operational cost and emissions of the two combustion modes. The DF-PCCI mode was more suitable than the CDC mode for the medium- to high-load operations, mainly due to the superior fuel economy, and low $NO_X$ and PM emissions. But the CDC was the optimal combustion mode at low loads because the DF-PCCI mode led to a deterioration in fuel consumption, low exhaust gas temperature, and high HC and CO emissions. Hutter et al. [31] analyzed the factors that reached the low-load limit in an NG-diesel DF-PCCI engine. Although the DF-PCCI mode has the advantage of having lower $CO_2$ emissions than CDC mode owing to the low carbon-to-hydrogen ratio of NG, the advantage disappeared as the engine load was decreased to the low-load limit. HC emissions also increased sharply. Throttling effectively reduced the HC emissions by increasing the equivalence ratio of the fuel-air charge. However, the $CO_2$ emissions increased due to the pumping loss in the throttled operation. Li et al. [32] performed experimental investigations in a six-cylinder NG-diesel DF-PCCI engine at 5.5 bar brake mean effective pressure (BMEP). In the premixed DF-PCCI combustion mode, which was realized by advancing the diesel SOI, there was higher thermal efficiency and lower HC emissions than in conventional diffusive dual-fuel combustion. In the DF-PCCI combustion mode, an advanced diesel SOI, a large EGR rate, and a moderate equivalence ratio were favorable for increasing thermal efficiency and reducing HC emissions. Gharehghani et al. [33] compared NG-biodiesel DF-PCCI combustion to NG-diesel DF-PCCI combustion in a single-cylinder engine at low loads. In both combustion modes, as the engine load was decreased, the HC and CO emissions increased. Superior thermal efficiency and emissions were achieved in the NG-biodiesel DF-PCCI mode than in the NG-diesel DF-PCCI mode because the higher oxygen content and cetane number in biodiesel improved the combustion quality and thus reduced the combustion loss.

Substantial studies have pointed out the barriers arising from low-load operations of DF-PCCI engines, namely, the deterioration in thermal efficiency, and HC and CO emissions. NG has lower reactivity than gasoline, and although this intensified the barriers at the low loads, very few studies addressed the causes and solutions for the barriers in NG-diesel DF-PCCI combustion mode. Therefore, the objective of this study is to investigate the causes of the barriers to the expansion of the low-load operating range in an NG-diesel DF-PCCI engine and to suggest a strategy for overcoming the barriers in terms of fuel-air mixture preparation. The experimental study was conducted in a heavy-duty compression-ignition engine. The engine was operated in the DF-PCCI mode fueled with NG and diesel over a low-load range from 0.3 to 0.6 MPa IMEP. Combustion and energy balance was analyzed to find the causes of the barriers and to evaluate the mixture

**Table 1**
Engine specifications.

| Item | Specification |
|---|---|
| Engine type | Single-cylinder, compression-ignition, direct-injection, naturally aspirated, heavy-duty |
| Combustion chamber shape | Re-entrant bowl |
| Displacement [L] | 0.981 |
| Bore [mm] | 100 |
| Stroke [mm] | 125 |
| Compression ratio [-] | 17.4 |
| Fuel injection equipment | Common-rail direct-injection system |
| EGR system | Cooled-EGR |

preparation strategy. A wide range of diesel SOI and the blending ratio of the two fuels was applied over the low loads to identify the barriers to the expansion of the low-load operating range and their causes. In response to the identified causes, a mixture preparation strategy was established and evaluated in terms of performance and emissions. Diesel SOI, a blending ratio of the NG and diesel, and an EGR rate were used to determine the optimization of the mixture preparation at each load.

## 2. Experimental setup and conditions

### 2.1. Experimental setup

A six-cylinder, heavy-duty, compression-ignition engine was modified to a single-cylinder engine for the experimental investigation with NG-diesel DF-PCCI combustion. The specifications of the engine and diesel injector are given in Tables 1 and 2, respectively. The engine was equipped with a common-rail, direct-injection system consisting of a high-pressure pump, a common-rail, and a solenoid-type injector. A cooled-EGR system, consisting of an EGR valve and an EGR cooler, was also installed. The EGR cooler was used to reduce the gas temperature in the downstream of the EGR system and to fix the air temperature at the intake port. The EGR rate in this study was obtained by measuring the concentrations of $CO_2$ in the exhaust gas and the intake air. The definition of the EGR rate is shown in Eq. (1).

$$EGR\ rate\,[\%] = \frac{CO_{2,intake} - CO_{2,ambient}}{CO_{2,exhaust} - CO_{2,ambient}} \times 100 \tag{1}$$

Tables 3 and 4 presents the properties of the diesel and NG implemented in this investigation, respectively. The previous study of the authors contains a detailed description of the fuels [34]. The experimental setup is illustrated in Fig. 1. The NG used as a low-reactivity fuel was introduced into the cylinder during the intake stroke to form a well-mixed NG-air mixture. An NG mass flow controller (Bronkhorst, F-202AV) and an air mass flow meter (Bronkhorst, F-106AI) were used to record the mass flow rates of the NG and air, respectively. The diesel used as a high-reactivity fuel was injected directly into the combustion chamber during the compression stroke, and it acted as an ignition source. A fuel consumption meter (AVL, 733S) was used to record the mass flow rate of the diesel. A piezoresistive type pressure sensor

**Table 2**
Diesel injector specifications.

| Item | Specification |
|---|---|
| Injector type | Solenoid injector |
| Number of holes | 8 |
| Nozzle hole diameter [mm] | 0.146 |
| Injection angle [°] | 146 |
| Hydraulic flow rate (HFR*) [$cm^3$/30 s] | 460 |
| Minimum injection quantity [mg/stroke] | 2 |

* HFR: diesel fuel volume flowing through an injector nozzle over a specified period of time

**Table 3**
Properties of diesel.

| Item | Value |
|---|---|
| Density at 293 K [kg/m3] | 826 |
| Cetane number [-] | 52.1 |
| Lower heating value [MJ/kg] | 42.5 |
| T10* (K) | 508 |
| T50* (K) | 562 |
| T90* (K) | 616 |
| Stoichiometric air-to-fuel ratio | 14.5 |

* T10, T50, T90: the temperature at which 10%, 50%, and 90% of fuel evaporates

**Table 4**
Properties of natural gas.

| Item | Value |
|---|---|
| Methane ($CH_4$) [%] | 91.31 |
| Ethane ($C_2H_6$) [%] | 5.34 |
| Propane ($C_3H_8$) [%] | 2.17 |
| I-Butane (i-$C_4H_{10}$) [%] | 0.46 |
| N-Butane (n-$C_4H_{10}$) [%] | 0.48 |
| I-Pentane (i-$C_5H_{10}$) [%] | 0.016 |
| N-Pentane (n-$C_5H_{10}$) [%] | 0.002 |
| Nitrogen ($N_2$) [%] | 0.222 |
| Motor octane number (MON) [-] | 124 |
| Methane number (MN) [-] | 81.7 |
| Lower heating value (LHV) [MJ/$Nm^3$] | 38.5 |
| Gas density [kg/$Nm^3$] | 0.829 |
| Stoichiometric air-to-fuel ratio | 16.9 |

(Kistler, 4045A5) and a piezoelectric type pressure sensor (Kistler, 6052C) were implemented to measure the intake and in-cylinder pressure, respectively. An exhaust gas analyzer (Horiba, Mexa-7100 DEGR) was used to determine the engine-out emissions, including $NO_X$, total HC (THC), CO, $CO_2$, and $O_2$. The $NO_X$ and THC concentrations in the exhaust gas were measured by chemiluminescence detection and flame ionization detection methods, respectively. A non-dispersive infrared method was used to measure the CO and $CO_2$ concentrations. Magneto-pneumatic detection was used to measure the $O_2$ concentration. A smoke meter (AVL, 415S) was used to determine the engine-out smoke emissions. The specifications of measurement equipment are presented in Table 5. A combustion analyzer (NI, cROI-9024) was used to record and process the measured values at a sampling rate of 0.2 crank angle degree (CAD). The pulse for each 0.2 CAD was generated by an optical rotary encoder (Autonics, E40S) installed on the crankshaft.

In this study, the NG substitution ratio (SR) was defined to represent the in-cylinder blending ratio between the NG and diesel. The NG SR was calculated based on the mass flow rate and lower heating value (LHV) of each fuel, as shown in Equation (2).

$$NG\ SR\,[\%] = \frac{\dot{m}_{NG} LHV_{NG}}{\dot{m}_{NG} LHV_{NG} + \dot{m}_{Diesel} LHV_{Diesel}} \times 100 \tag{2}$$

where $\dot{m}_{NG}$ and $\dot{m}_{Diesel}$ are the mass flow rates of the NG and diesel, respectively. $LHV_{NG}$ and $LHV_{Diesel}$ are the LHVs of the NG and diesel, respectively. The equivalence ratio of the DF-PCCI concept was calculated by Equation (3) [35].

$$Equivalence\ ratio\,[-] = \frac{\dot{m}_{NG} AFR_{NG} + \dot{m}_{Diesel} AFR_{Diesel}}{\dot{m}_{air}} \tag{3}$$

where $\dot{m}_{air}$ is the mass flow rate of the air. $AFR_{NG}$ and $AFR_{Diesel}$ are the stoichiometric air-to-fuel ratios of the NG and diesel, respectively.

### 2.2. Analysis of combustion and energy balance

In this study, an analysis of combustion and energy balance was conducted to understand the combustion phenomena occurring in the

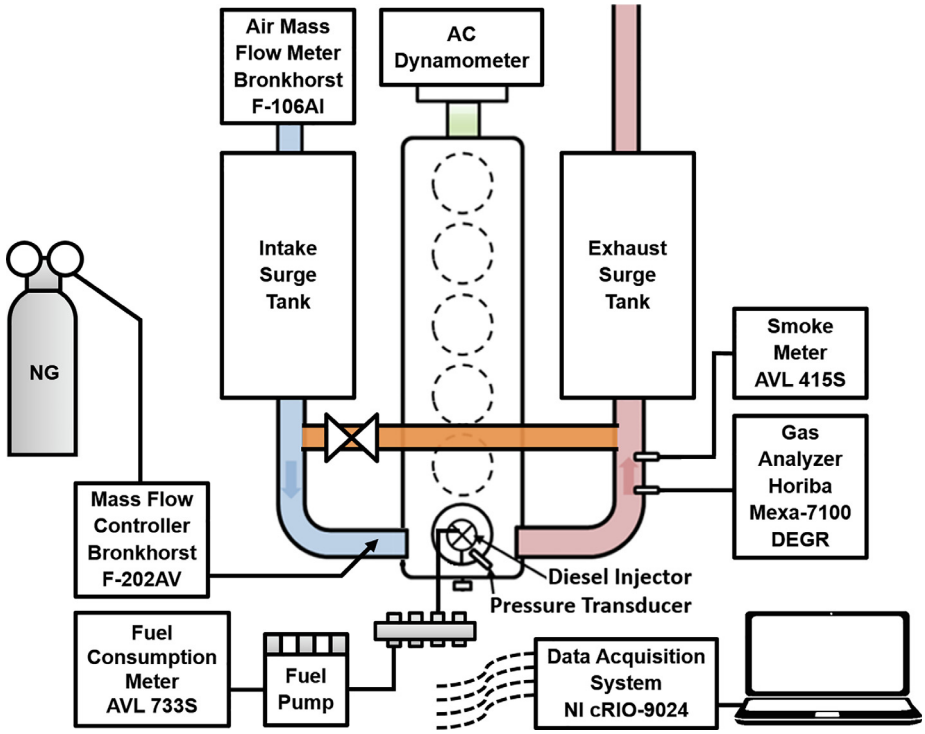


**Fig. 1.** Schematic diagram of engine setup [34]

**Table 5**
Specifications of measurement equipment.

| Measured parameters | Device | Measurement range | Linearity/ accuracy | Repeatability |
|---|---|---|---|---|
| CO | HORIBA | 0–10 vol% | ≤ ± 1% of FS* | ≤ ± 0.5% of FS* |
| $CO_2$ | MEXA-7100 | 0–16 vol% | | |
| $NO_X$ | DEGR | 0–10,000 ppm | | |
| THC | | 0–20,000 ppm | | |
| $O_2$ | | 0–25 vol% | | |
| PM | AVL Smoke Meter 415S | 0–10 FSN | – | ≤ ± 0.005 FSN ≤3% of MV** |
| Diesel mass flow rate | AVL 733S | 0–150 kg/h | ≤ ± 0.12% of MV** | – |
| NG mass flow rate | Bronkhorst F-202AV | 0.08–4 kg/h | ≤ ± 0.5% of MV** ≤ ± 0.1% of FS* | ≤ ± 0.2% of MV** |
| Air mass flow rate | Bronkhorst F-106AI | 5–125 kg/h | ≤ ± 0.1% of FS* | ≤ ± 0.2% of MV** |
| In-cylinder pressure | Kistler 6056A | 0–25 MPa | ≤ ± 0.3% of FS* | – |
| Intake pressure | Kistler 4045A5 | 0–0.5 MPa | ≤ ± 0.3% of FS* | – |

* FS: full scale
** MV: measurement value

DF-PCCI engine and, thus, to assess the efficiency of the energy conversion in the engine.

The measured pressure traces were used to acquire the net HRR [36]. The in-cylinder pressure profile for the HRR analysis was acquired through ensemble averaging the pressure data of 100 consecutive cycles for each set of experimental conditions. A smoothing equation was applied to the in-cylinder pressure profile because the first derivative of pressure with respect to CAD was noisy [37]. The smoothing seems to offer a logical compromise between tracking meaningful pressure signal information and avoiding the noisy derivative term [38]. Combustion parameters, such as CA10, CA50, and CA90, were calculated to quantify combustion phasing and combustion duration. The CA10, CA50, and CA90 were defined as the crank angle where 10%, 50%, and 90% of the total (cumulative) heat is released, respectively. CA50 represented the combustion phasing, and the combustion duration was determined by the difference between CA90 and CA10.

Degree of constant volume (DOCV) was used to quantify the heat release characteristic of the DF-PCCI combustion. In order to obtain high thermal efficiency in internal combustion engines, the heat is released for a short combustion duration near the TDC [6,39]. The DOCV is a quantification of the constant-volume combustion ratio in actual engine combustion compared to the theoretical Otto cycle, as shown in Eq. (4) [40].

$$Degree\ of\ constant\ volume = \frac{1}{\eta_{Otto} Q} \int \left\{ 1 - \left[ \frac{V_d + V_c}{V(\theta)} \right]^{1-\gamma} \right\} \frac{dQ}{d\theta} d\theta \tag{4}$$

where $\eta_{Otto}$ is the theoretical efficiency of Otto-cycle, and Q is the cumulative heat release. $V_d$ and $V_c$ are the displacement and clearance volume, respectively. $dQ/d\theta$ is the HRR.

The fuel energy in internal combustion engines consists of gross indicated power, combustion loss, heat transfer loss, and exhaust loss [19,41,42]. The gross indicated power is divided into net indicated power and pumping loss [42]. Equations (5) and (6) were used to calculate the indicated thermal efficiency (ITE) and combustion loss, respectively. The previous study of the authors contains detailed descriptions of the equations [34].

$$ITE = \frac{3.6P}{\dot{m}_{NG} LHV_{NG} + \dot{m}_{Diesel} LHV_{Diesel}} \tag{5}$$

where P is the net indicated power [kW] and calculated through the indicated work and the engine speed.

$$Combustion\ loss = \frac{\dot{m}_{CO} LHV_{CO} + \dot{m}_{THC} LHV_{Fuel}}{\dot{m}_{NG} LHV_{NG} + \dot{m}_{Diesel} LHV_{Diesel}} \tag{6}$$

where $\dot{m}_{CO}$ and $\dot{m}_{THC}$ are the mass flow rates of the CO and THC, respectively. The mass flow rates of THC and CO were determined by multiplying the mass flow rate of exhaust gas by the mass fractions of

THC and CO, respectively. The exhaust loss was calculated using the measured exhaust gas temperature and intake temperature, as shown in Eq. (7) [19].

$$Exhaust\ loss = \frac{\dot{m}_{exh}[h_{exh}(T_{exh}) - h_{int}(T_{int})]}{\dot{m}_{NG}LHV_{NG} + \dot{m}_{Diesel}LHV_{Diesel}} \quad (7)$$

where $\dot{m}_{exh}$ is the mass flow rate of the exhaust. $h_{exh}$ and $h_{int}$ are the exhaust and intake enthalpies, respectively. The specific enthalpies were acquired using the NIST reference fluid thermodynamic and transport properties [43]. It was assumed that the exhaust consists only of $CO_2$, $H_2O$, $O_2$, $N_2$, NO, HC, and CO. The heat transfer loss was calculated by subtracting the gross indicated power, combustion loss, and exhaust loss from the fuel energy [41,42].

### 2.3. Experimental conditions and procedure

Previous studies related to NG-diesel DF-PCCI combustion have shown that $NO_X$ and PM emissions effectively reduced by modulating diesel SOI and NG SR in low-load operations [28–31]. However, the deterioration in thermal efficiency, and HC and CO emissions were recognized as the barriers to the expansion of the low-load limit.

In this study, the causes of the barriers were analyzed, and a mixture preparation strategy was proposed to overcome the barriers in an NG-diesel DF-PCCI engine. Firstly, baseline DF-PCCI points were determined by combinations of the diesel SOI and NG SR to find the causes of the barriers. Combustion and energy balance were analyzed to investigate the causes of the barriers. Then the mixture preparation strategy was established based on the causes. The optimal points found in the strategy were compared to the baseline DF-PCCI points in terms of thermal efficiency and HC and CO emissions.

The experimental conditions for the investigation are shown in Table 6. The engine speed was fixed as 1400 rpm. The engine load was varied from 0.3 to 0.6 MPa net IMEP, which is a range of low-load operations. Other parameters, such as intake pressure, intake temperature, coolant temperature, and diesel injection pressure, were fixed. The intake pressure was maintained at 0.1 MPa during the experiments. In general, the increase of the intake pressure under low loads is considered to be difficult due to the low enthalpy (the low exhaust gas temperature and low flow rate) of the exhaust gas. The reduction of the exhaust gas temperature due to the increase of boost pressure also leads to low conversion efficiency of oxidation catalysts in low-load operations of DF-PCCI engines. This fact can explain the validity of the naturally aspirated conditions utilized in the current study. The coolant temperature was controlled by the value at the outlet of the engine block and head.

It should be noted that there were constraints during the experiment. The $NO_X$ and PM emissions were set lower than the EURO VI regulation ($NO_X$ less than 0.4 g/kWh, PM less than 0.01 g/kWh) because DF-PCCI concepts were proposed to reduce the $NO_X$ and PM emissions from CDC. The coefficient of variation (COV) of IMEP was kept at less than 5% to maintain stable combustion. To prevent engine damage, the MPRR was kept at less than 1 MPa/CAD [44,45]. Each set of the experimental conditions was repeated three times, and the exhaust gas emission and fuel mass flow rate measurements were taken again three times. The coefficient of variation (standard deviation divided by the mean value) for the emission and mass flow rate measurements were less than 3% and 2%, respectively, demonstrating satisfactory repeatability [46].

**Table 6**
Experimental conditions.

| Item | Value |
|---|---|
| Engine load (net IMEP) [MPa] | 0.3–0.6 |
| Engine speed [rpm] | 1400 |
| Intake pressure [MPa] | 0.1 |
| Intake temperature [K] | 298 ± 1 |
| Coolant temperature [K] | 353 ± 1 |
| Diesel injection pressure [MPa] | 100 |

## 3. Results and discussion

### 3.1. Barriers for expanding low-load operating range

In this section, the baseline DF-PCCI points were selected to investigate barriers and their causes in low-load operations of NG-diesel DF-PCCI engines. The baseline DF-PCCI points were determined based on high ITE, and the $NO_X$ and PM emissions below the constraints. Therefore, CA50 was chosen as a control parameter to determine the baseline DF-PCCI points at the low-load range from 0.3 to 0.6 MPa IMEP. The CA50 is a crucial factor influencing the thermal efficiency and emissions in internal combustion engines [6,39]. For example, when the CA50 is located before TDC, the thermal efficiency is reduced because of the negative work, and the high in-cylinder temperature causes the $NO_X$ emissions to increase. Conversely, when the CA50 is retarded well beyond an optimal location, the thermal efficiency is reduced, and HC and CO emissions increase due to a reduction in the combustion efficiency. In this paper, only the experimental results of the thermal efficiency and emissions at 0.45 MPa IMEP were presented. This is because the trends of the results were similar at the low-load range from 0.3 to 0.6 MPa IMEP, although there were differences in specific values.

Fig. 2 shows the NG SR and ITE according to the CA50 and diesel SOI. Increasing the NG SR at the fixed diesel SOI caused the CA50 to be retarded. As the diesel SOI was retarded toward TDC, the NG SR was increased to keep the CA50 constant. In other words, the CA50 was retarded by increasing the NG SR and advancing the diesel SOI. The increase in the NG SR formed a well-premixed fuel-air mixture because the fraction of the fuel introduced during the intake stroke increased [14,47]. The mixing between the fuel and air was also increased by the advance in the diesel SOI [14,17]. Therefore, the increase of the NG SR and the advance of the diesel SOI caused the increase in the homogeneity of the mixture and the reduction in its local reactivity, finally retarding the CA50 [34]. There was an optimal ITE region consisting of a combination of CA50 and diesel SOI at each low-load. Increasing the engine load caused the CA50 location with the optimal ITE region to be retarded. The optimal CA50 for the best ITE at 0.3 MPa IMEP was located around −2 CAD after top dead center (aTDC). When CA50 was located before TDC, the negative work generally reduced the thermal efficiency [6,39]. However, retarding the CA50 after TDC at 0.3 MPa IMEP caused the combustion efficiency to deteriorate abruptly, which resulted in low ITE [34]. The optimal CA50 for the best ITE was located around 2 and 4 CAD aTDC at 0.45 and 0.6 MPa IMEP, respectively.

Fig. 3 shows the $NO_X$ and PM emissions according to the CA50 and diesel SOI at 0.45 MPa IMEP. The $NO_X$ and PM emissions were reduced when the CA50 was retarded and the diesel SOI was advanced. The advance in the diesel SOI increased the homogeneity of the mixture. The retardation of the CA50 resulted from a higher NG SR. The higher NG SR also increased the homogeneity of the mixture. Therefore, the increase in the mixture homogeneity, which lowered the local reactivity of the mixture, reduced the combustion temperature and, thus, $NO_X$ and PM emissions. At all combinations of the CA50 and diesel SOI tested, the PM emissions were below the constraints.

Fig. 4 shows the THC and CO emissions according to the CA50 and diesel SOI at 0.45 MPa IMEP. The NG SR strongly affected THC emissions. It is noted that the trend in THC emissions in Fig. 4 was similar to that in the NG SR in Fig. 2. The higher THC emissions with the higher NG SR were due to the low combustion temperature and the NG trapped in the crevices during the compression stroke [34]. The CO emissions increased when the CA50 was retarded and the diesel SOI was advanced. The high CO emissions were due to the low combustion temperature of the more homogeneous fuel-air mixture. The CO

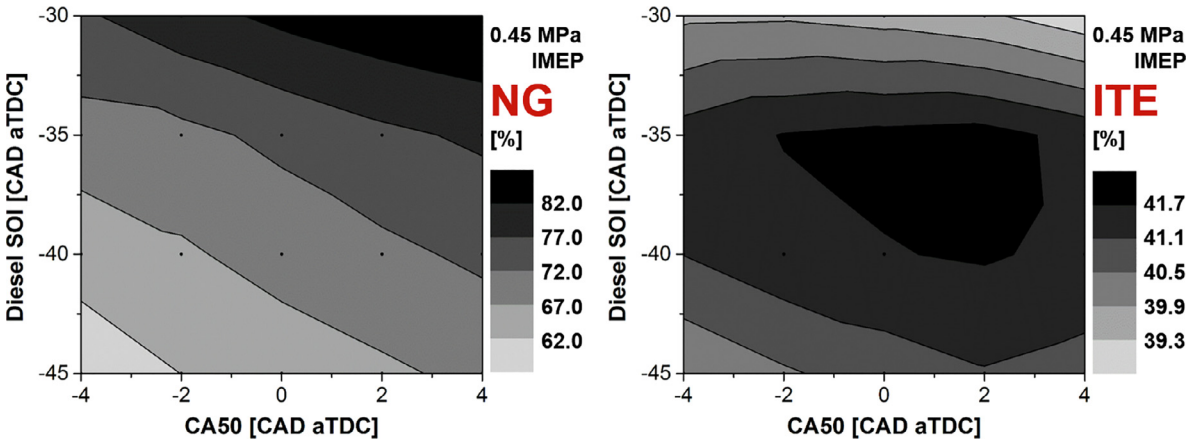


**Fig. 2.** Natural gas substitution ratio (left) and indicated thermal efficiency (right) according to CA50 and diesel SOI timing at 0.45 MPa IMEP for determining the baseline case.

emissions were more affected by the CA50 than by the diesel SOI. The baseline DF-PCCI point at 0.45 MPa IMEP was selected as the CA50 of 2 CAD aTDC and the diesel SOI of −40 CAD aTDC because of the high ITE, and the $NO_X$ and PM emissions below the constraints. The procedures for selecting the baseline DF-PCCI points at 0.3 and 0.6 MPa IMEP were the same for those at 0.45 MPa IMEP.

Table 7 shows the engine operating conditions for the selected baseline DF-PCCI points over the low-load range from 0.3 to 0.6 MPa IMEP. The optimal CA50, and low $NO_X$ and PM emissions were acquired with the combination of diesel SOI and NG SR at each low-load of the DF-PCCI engine. As the engine load was increased, an advancement of diesel SOI and a higher NG SR should be applied to retard the CA50 and to reduce the $NO_X$ emissions [19,28].

Fig. 5 shows the in-cylinder pressure and HRR for the baseline DF-PCCI points. The corresponding performance and emissions are shown in Fig. 6. The optimal CA50 was retarded as the engine load was increased. It is noted that although the best ITE was located around the CA50 of 4 CAD aTDC at 0.6 MPa IMEP, the optimal CA50 was selected as 6 CAD aTDC in order to reduce the $NO_X$ emissions below the constraint. The combination of the diesel SOI and NG SR alone achieved low $NO_X$ and PM emissions that met the constraints at all the low loads investigated. However, the THC and CO emissions increased rapidly as the engine load was decreased. In each low-load operation of the NG-diesel DF-PCCI engine, the increase in the homogeneity of the mixture contributed to reducing the $NO_X$ and PM emissions simultaneously, but it also increased the THC and CO emissions, because the increase of the mixture homogeneity reduced the combustion temperature [48]. In this situation, as the engine load was decreased, an overly-lean fuel-air mixture was formed. It is noted that the equivalence ratio of the mixture was reduced from 0.5 at 0.6 MPa IMEP to 0.29 at 0.3 MPa IMEP, as shown in Table 7. At the lowest engine load, the overly-lean mixture caused the high THC and CO emissions by intensifying the effects of increasing the mixture homogeneity. Unburned methane accounts for a significant portion of THC emissions in NG-diesel dual fuel engines [49]. The unburned methane emissions are formed in regions of the overly-lean mixture where the combustion temperature is low [50]. High CO can be formed in the overly-lean mixture regions in the temperature below 1450 K [50,51]. The CO was not oxidized to $CO_2$ because the combustion temperature was low. Therefore, there was an abrupt increase in THC and CO emissions at 0.3 MPa IMEP.

As the engine load was decreased, the ITE also deteriorated significantly. In order to determine the reason for the low ITE, an analysis of energy balance was conducted. Fig. 7 shows the energy balance for the baseline DF-PCCI points at the low-load range of 0.3 to 0.6 MPa IMEP. As the engine load was decreased, the combustion and heat transfer losses increased. However, the exhaust loss was reduced at the lower engine load, which offset the increase in the heat transfer loss [19]. The pumping loss was slightly reduced with decreasing the engine load. Therefore, the higher combustion loss was considered as the main cause of the low ITE. As the engine load was moved from 0.6 to 0.3 MPa IMEP, the combustion loss increased from 3.8% to 8.7%. The high combustion loss at 0.3 MPa IMEP was due to the low combustion temperature of the overly-lean mixture, which had the equivalence ratio of 0.29.

The barriers and their causes for expanding the low-load operating range were investigated by selecting the baseline DF-PCCI points at the

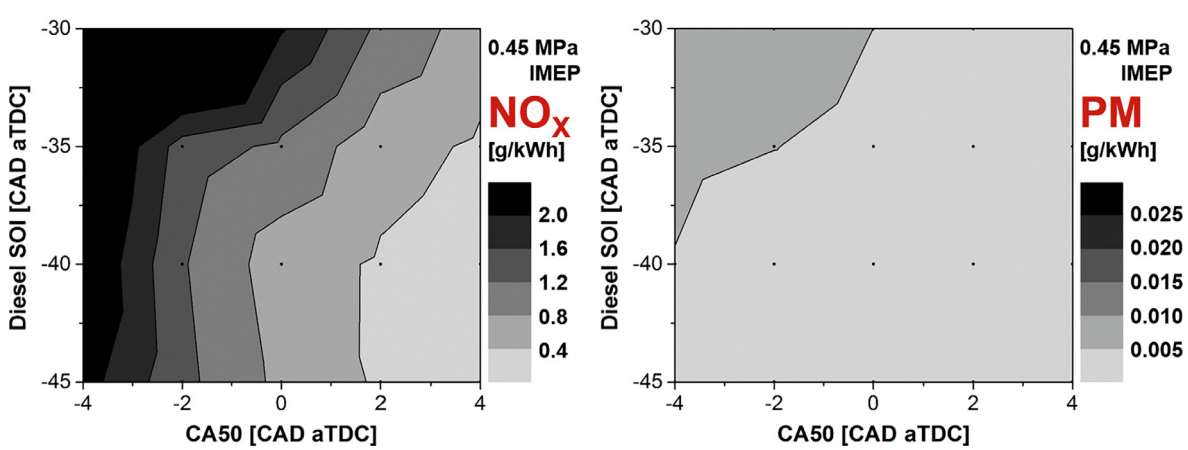


**Fig. 3.** $NO_X$ (left) and PM (right) emissions according to CA50 and diesel SOI timing at 0.45 MPa IMEP for determining the baseline case.

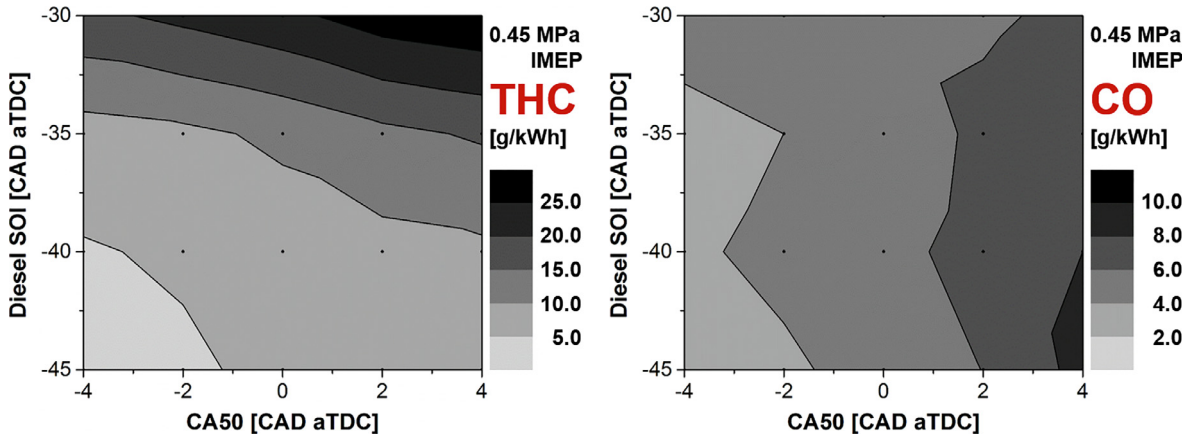


**Fig. 4.** THC (left) and CO (right) emissions according to CA50 and diesel SOI timing at 0.45 MPa IMEP for determining the baseline case.

**Table 7**
Engine operating conditions for baseline DF-PCCI.

| IMEP [MPa] | 0.3 | 0.45 | 0.6 |
|---|---|---|---|
| NG flow rate [kg/h] | 0.39 | 0.66 | 0.91 |
| Diesel flow rate [kg/h] | 0.36 | 0.3 | 0.32 |
| Air flow rate [kg/h] | 40.6 | 40.4 | 40.3 |
| Diesel SOI [CAD aTDC] | −35 | −40 | −55 |
| NG substitution [%] | 54 | 71 | 76 |
| EGR rate [%] | 0 | 0 | 0 |
| Equivalence ratio [-] | 0.29 | 0.38 | 0.5 |

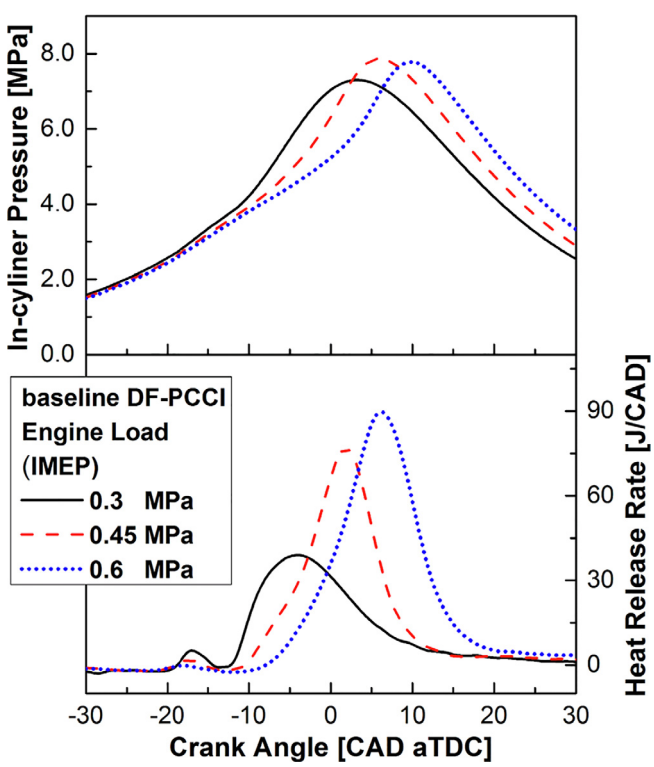


**Fig. 5.** In-cylinder pressure and heat release rate for baseline DF-PCCI over a low-load range from 0.3 to 0.6 MPa IMEP.

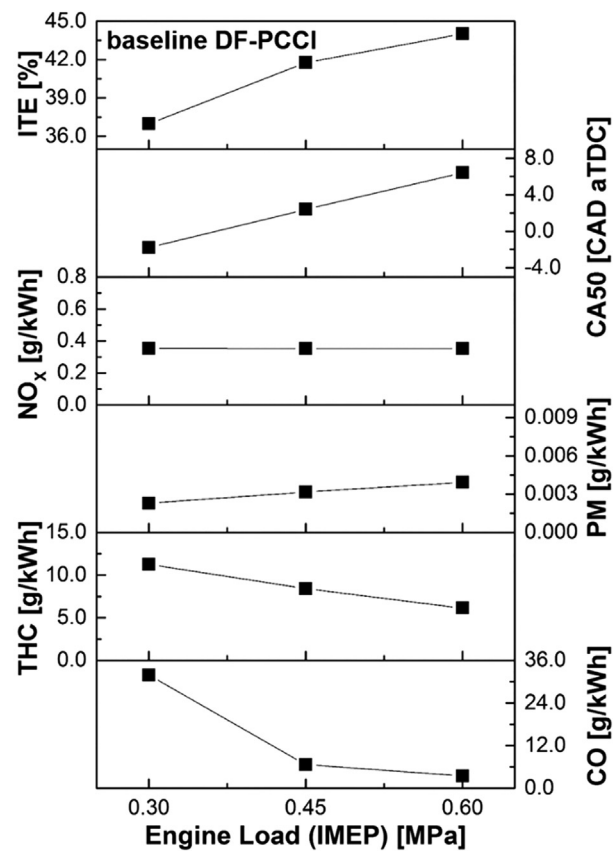


**Fig. 6.** Performance and emissions for baseline DF-PCCI over a low-load range from 0.3 to 0.6 MPa IMEP.

low-load range from 0.3 to 0.6 MPa IMEP. The barriers were identified as low ITE and high THC and CO emissions. The main cause of the barriers can be summarized as the formation of the overly-lean mixture at the lowest engine load. The overly-lean mixture increased the mixture homogeneity and thus reduced the combustion temperature and the combustion efficiency. Therefore, the ITE, and THC and CO emissions deteriorated significantly.

### 3.2. Mixture stratification for overcoming barriers at low loads

In the previous section, the formation of the overly-lean mixture was identified as the main cause of low ITE, and high THC and CO emissions at 0.3 MPa in the NG-diesel DF-PCCI engine. Therefore, an increase in the local equivalence ratio of the mixture can improve the ITE, and the THC and CO emissions. A higher local equivalence ratio is achieved by retarding the diesel SOI and reducing the NG SR in DF-PCCI concepts [34,47]. In other words, mixture stratification increases the equivalence ratio and reactivity of the local mixture where the

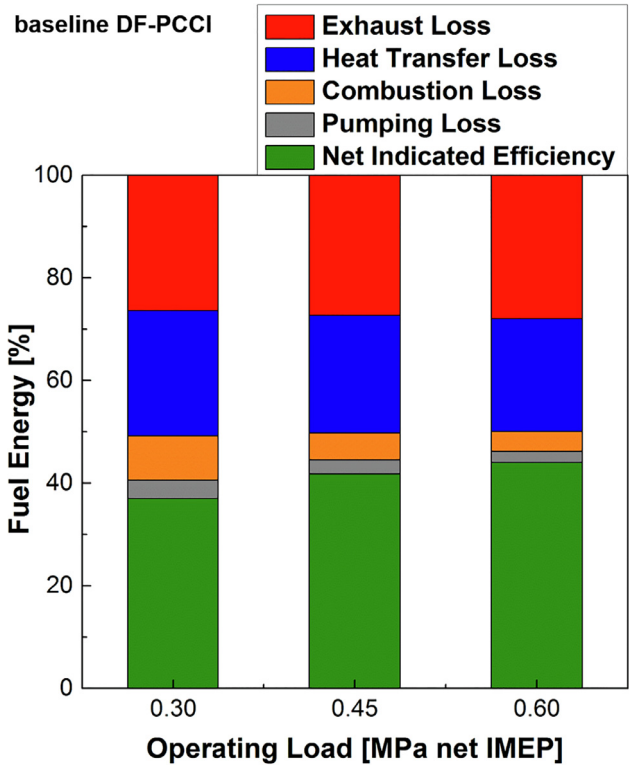


**Fig. 7.** Energy balance for baseline DF-PCCI over a low-load range from 0.3 to 0.6 MPa IMEP.

combustion initiates. In fact, THC and CO were reduced with the mixture stratification, as shown in Fig. 4. However, although the mixture stratification contributed to reducing THC and CO emissions, $NO_X$ emissions increased, as shown in Fig. 3, and the CA50 was advanced before TDC, as shown in Fig. 2. Therefore, cooled-EGR was implemented with the mixture stratification to reduce the $NO_X$ emissions and to retard the CA50.

Fig. 8 shows the HRR, CA50, and $NO_X$ emissions according to the diesel SOI and EGR rate at 0.3 MPa IMEP. When the diesel SOI was retarded from −35 to −25 CAD aTDC at NG SR of 50%, without the implementation of the EGR rate, the CA50 was advanced and the peak HRR was increased. The higher peak HRR resulted in higher $NO_X$ emissions. In this situation, as the EGR rate of 40% was introduced, the CA50 was retarded and the $NO_X$ emissions decreased because of the intake dilution effect of the EGR [52]. As a result, after the introduction of the EGR, the CA50 and $NO_X$ emissions had values similar to those in the initial state. Fig. 9 shows the HRR, CA50, and $NO_X$ emissions according to the NG SR and EGR rate at 0.3 MPa IMEP. When the NG SR was reduced from 60% to 40% at diesel SOI of −30 CAD aTDC without the EGR, the CA50 advanced, and the peak HRR increased. The effects were similar to those of the retardation of the diesel SOI. When EGR was introduced, the CA50 and $NO_X$ emissions returned to their initial levels. In summary, although a retardation of diesel SOI and a lower NG SR effectively reduced the THC and CO emissions, the CA50 advanced, and the $NO_X$ emissions increased. However, the implementation of EGR effectively offset the advanced CA50 and the increased $NO_X$ emissions [53]. The effect of EGR rate on the CA50 and the $NO_X$ emissions in the DF-PCCI engine was similar at the low loads investigated.

Table 8 shows the engine operating conditions for the mixture stratification strategy at 0.3 MPa IMEP. The baseline DF-PCCI corresponds to the selected point in the previous section. In the previous study comparing diesel PCCI and gasoline-diesel DF-PCCI at an extremely low load condition, the HC and CO emissions as well as the fuel consumption were much lower in the diesel PCCI than in the DF-PCCI [21]. Therefore, the diesel PCCI case was also tested at 0.3 MPa IMEP. There was a gradual increase in the degree of mixture stratification until it reached the baseline DF-PCCI, Strategy 1, Strategy 2, and diesel PCCI. In other words, the diesel SOI was retarded from −35 at the baseline DF-PCCI to −25 CAD aTDC at the Strategy 2. The NG SR was reduced from 54% at the baseline DF-PCCI to 23% at the Strategy 2. The EGR rate was increased from 0% at the baseline DF-PCCI to 50% at the Strategy 2. A higher EGR rate was implemented in the diesel PCCI (NG SR = 0%) than in the Strategy 2 of the DF-PCCI. Fig. 10 shows the performance and emissions of the baseline DF-PCCI, Strategy 1, Strategy 2, and diesel PCCI. The CA50 applied in the Strategy 2 and diesel PCCI was more advanced than in the baseline DF-PCCI and Strategy 1 to reduce the THC and CO emissions. As shown in Fig. 4, an advanced CA50 effectively reduced the THC and CO emissions. Despite the advanced CA50, the Strategy 2 and diesel PCCI had the $NO_X$ emissions levels similar to those of the baseline DF-PCCI and Strategy 1, owing to the higher EGR rate. Low $NO_X$ and PM emissions, which met the constraints, were seen in all cases.

The ITE improved when the degree of mixture stratification was increased. The higher ITE was due to the improvement in the combustion performance. Fig. 11 shows the HRR and combustion parameters according to the mixture stratification. Figs. 12 and 13 show the pressure-volume curves and DOCV according to the mixture stratification, respectively. As shown in Fig. 11, as the degree of mixture stratification was increased in the DF-PCCI, the HRR profile peak grew higher, its combustion duration became narrower. Although the local reactivity of the mixture in the Strategy 2 was higher than those in the

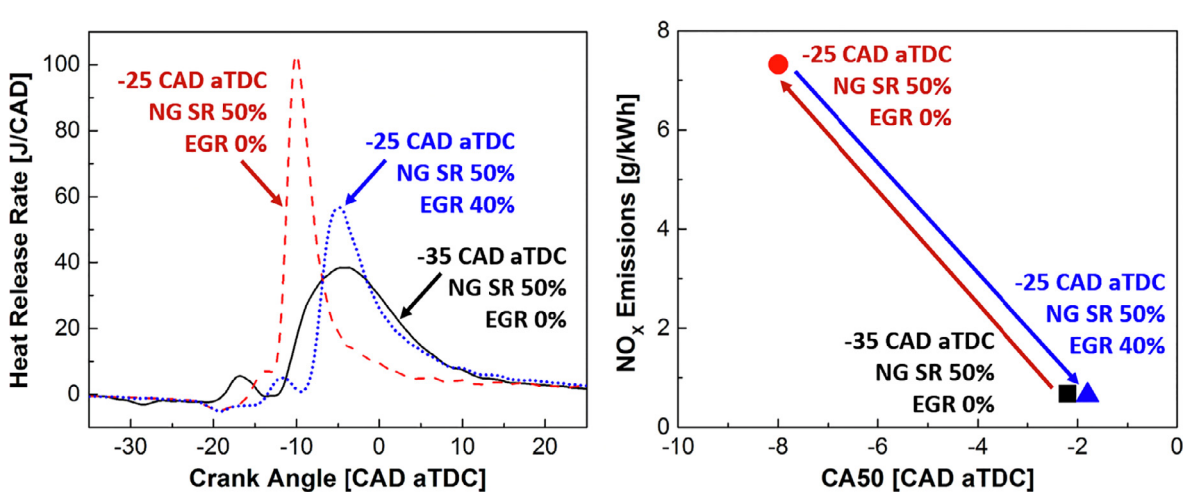


**Fig. 8.** Heat release rate (left) and CA50 and $NO_X$ emissions (right) according to diesel SOI timing and EGR rate at 0.3 MPa IMEP.

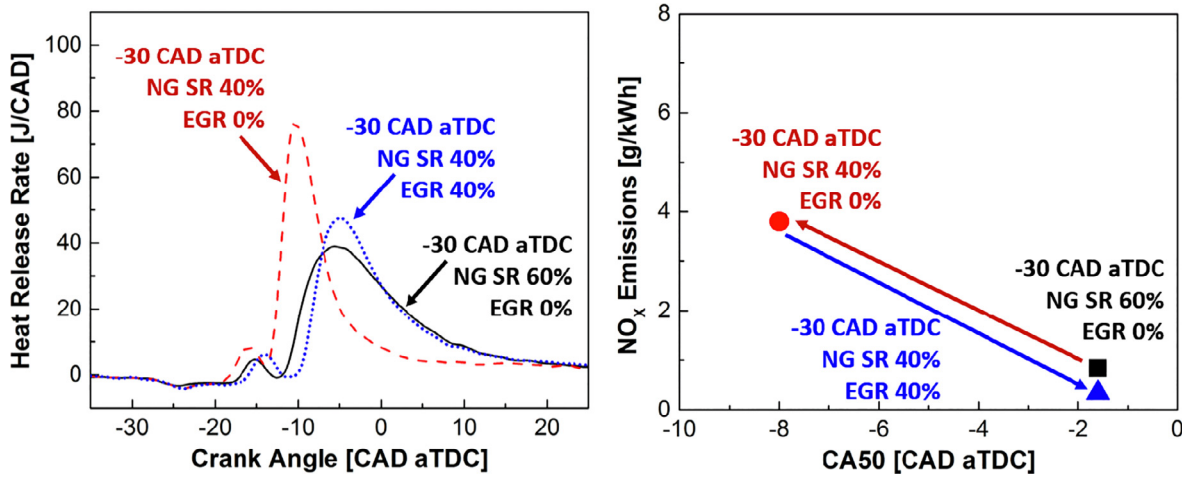


**Fig. 9.** Heat release rate (left) and CA50 and $NO_X$ emissions (right) according to natural gas substitution ratio and EGR rate at 0.3 MPa IMEP.

**Table 8**
Engine operating conditions at 0.3 MPa IMEP for evaluating a mixture stratification strategy.

| net IMEP [MPa] | baseline | Strategy 1 | Strategy 2 | Diesel PCCI |
|---|---|---|---|---|
| NG flow rate [kg/h] | 0.39 | 0.26 | 0.16 | 0 |
| Diesel flow rate [kg/h] | 0.36 | 0.47 | 0.57 | 0.73 |
| Air flow rate [kg/h] | 40.6 | 24.3 | 20.3 | 17.7 |
| Diesel SOI [CAD aTDC] | −35 | −30 | −25 | −25 |
| NG substitution [%] | 54 | 38 | 23 | 0 |
| EGR rate [%] | 0 | 40 | 50 | 57 |
| Equivalence ratio [-] | 0.29 | 0.46 | 0.54 | 0.60 |

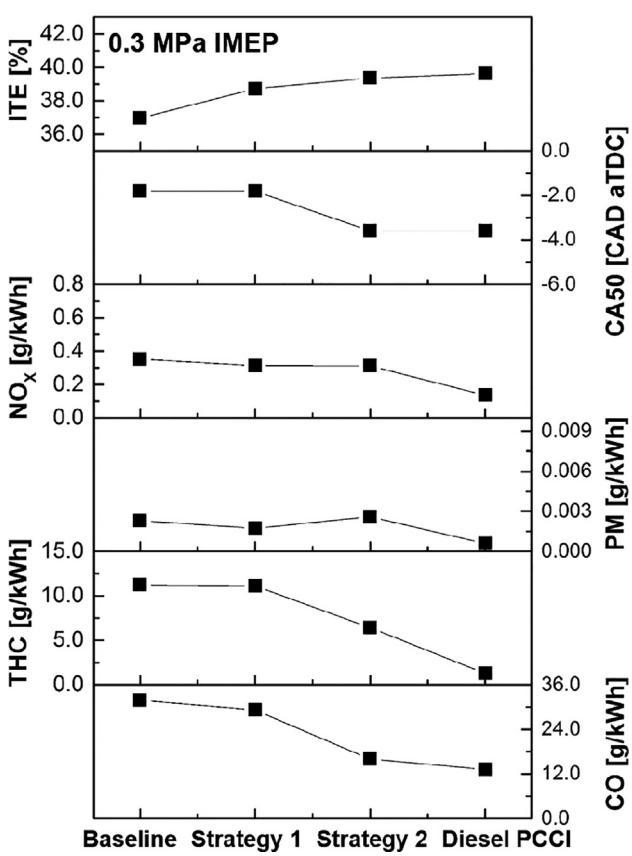


**Fig. 10.** Performance and emissions according to mixture stratification at 0.3 MPa IMEP.

baseline DF-PCCI, the combustion initiation was more retarded because of the higher EGR rate. The higher reactivity in the Strategy 2 enhanced the combustion, which resulted in the higher peak HRR and an earlier end of the combustion. Therefore, the pressure in the Strategy 2 had lower values than those in the baseline DF-PCCI before the combustion initiation, mainly because of the thermal effect of EGR, and it increased abruptly with the ignition of the higher mixture reactivity, as shown in Fig. 12. This caused the DOCV to increase from 0.91 at the baseline DF-PCCI to 0.94 at the Strategy 2, as shown in Fig. 13. In other words, the mixture stratification strategy caused the combustion to approach the theoretical Otto-cycle by controlling the HRR [54]. Therefore, the ITE increased in the DF-PCCI as the degree of mixture stratification was increased at 0.3 MPa IMEP. The ITE was more improved in the diesel PCCI than in the Strategy 2 of the DF-PCCI. The higher ITE was due to a higher peak HRR and a narrower combustion duration, as shown in Fig. 11.

The mixture stratification strategy also effectively reduced the THC and CO emissions. The reduction in the THC and CO emissions was mainly due to the enhanced combustion performance and the reduced mass flow rate of NG. The increase in the degree of mixture stratification increased the local reactivity of the mixture and thus enhanced the combustion performance. Unburned NG was the main source of THC emissions, and an increase of NG SR favored CO formation, as discussed in the previous section. Therefore, the lower NG SR with the mixture stratification strategy contributed to reducing the THC and CO formations.

The mixture stratification strategy was also implemented and assessed at 0.45 and 0.6 MPa IMEP. The engine operating conditions for DF-PCCI with mixture stratification at the low-load range from 0.3 to 0.6 MPa IMEP are shown in Table 9. A retardation of diesel SOI, a lower NG SR, and a higher EGR rate were implemented in the mixture stratification strategy than in the baseline DF-PCCI. As the engine load was increased, the required EGR rate for realizing the mixture stratification strategy was reduced mainly because of the higher equivalence ratio.

Fig. 14 shows the in-cylinder pressure and HRR for the DF-PCCI with mixture stratification. The corresponding performance and emissions are compared to those of the baseline DF-PCCI in Fig. 15. The $NO_X$ and PM emissions were below the constraints in the baseline DF-PCCI and DF-PCCI with mixture stratification. An advanced CA50 was applied to the DF-PCCI with mixture stratification in order to reduce the THC and CO emissions. The THC and CO emissions were reduced with a higher degree of mixture stratification at the low loads. The principles of reducing the THC and CO emissions were explained by the enhanced combustion performance and the reduced mass flow rate of NG, as discussed in the previous paragraph. The reduction in the THC and CO emissions by the mixture stratification was greater at the lowest load.

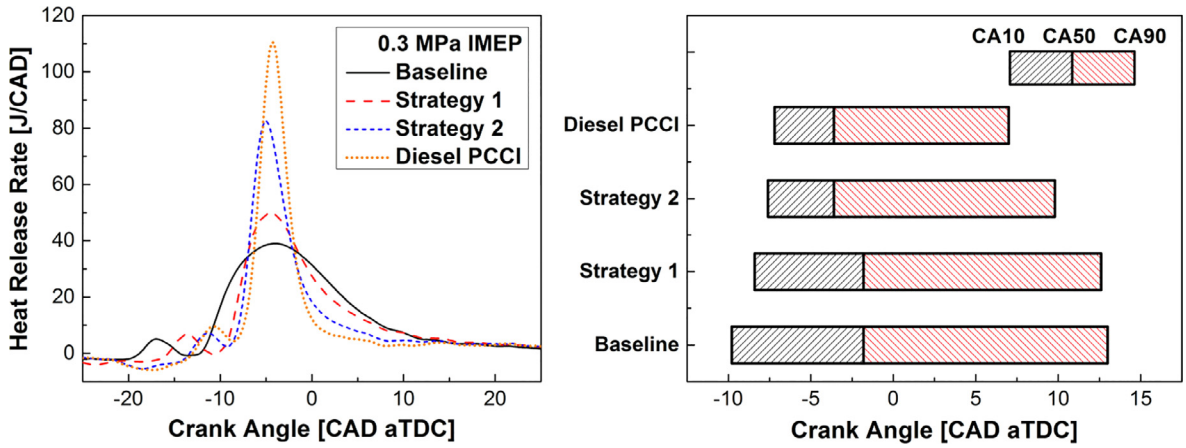


**Fig. 11.** Heat release rate (left) and combustion parameters (right) according to mixture stratification at 0.3 MPa IMEP.

The DF-PCCI with mixture stratification reduced the THC and CO emissions by about 43% and 50% at 0.3 MPa IMEP, respectively, compared to the baseline DF-PCCI.

The ITE was also improved by the mixture stratification strategy at the low loads investigated. The ITE improvement was also maximized at the lowest load. A more specific analysis of the ITE is shown in Fig. 16, which depicts the energy balance for the baseline DF-PCCI and DF-PCCI with mixture stratification at the low loads. Despite the higher heat transfer losses, the DF-PCCI with mixture stratification had lower exhaust and combustion losses than the baseline DF-PCCI. The reduction in the exhaust losses was due to more efficient work conversion, as shown in Fig. 12. Enhanced combustion performance, reduced mass flow rates of NG, and advanced CA50 all account for the reduction in the combustion losses. The enhanced combustion performance and advanced CA50 effectively reduced the combustion losses but increased the heat transfer losses. The main contributor to the improvement in the ITE at the low loads was the reduction in the combustion losses. Although the reduction in the combustion losses was higher than the increase in the ITE, the contribution of the combustion losses was partially offset by the increase in the heat transfer losses. The higher improvement in the ITE at the lowest load was due to the higher reduction in the combustion loss.

Diesel PCCI was implemented at 0.45 MPa IMEP because it effectively improved the ITE at 0.3 MPa IMEP. Fig. 17 shows the energy balance for the DF-PCCI with mixture stratification and diesel PCCI at 0.3 and 0.45 MPa IMEP. The application of the diesel PCCI at 0.6 MPa IMEP was limited by requiring excessive EGR rates in this study. Although the diesel PCCI was effective in improving the ITE at 0.3 MPa IMEP, the ITE became worse in the diesel PCCI than in the DF-PCCI at 0.45 MPa IMEP. While the combustion efficiency gain due to the application of the diesel PCCI was higher than the increase in the heat transfer loss at 0.3 MPa IMEP, the gain did not exceed the increase in

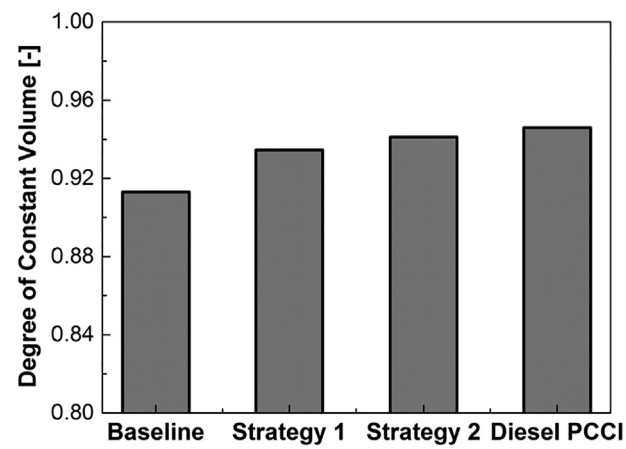


**Fig. 13.** Degree of constant volume according to mixture stratification at 0.3 MPa IMEP.

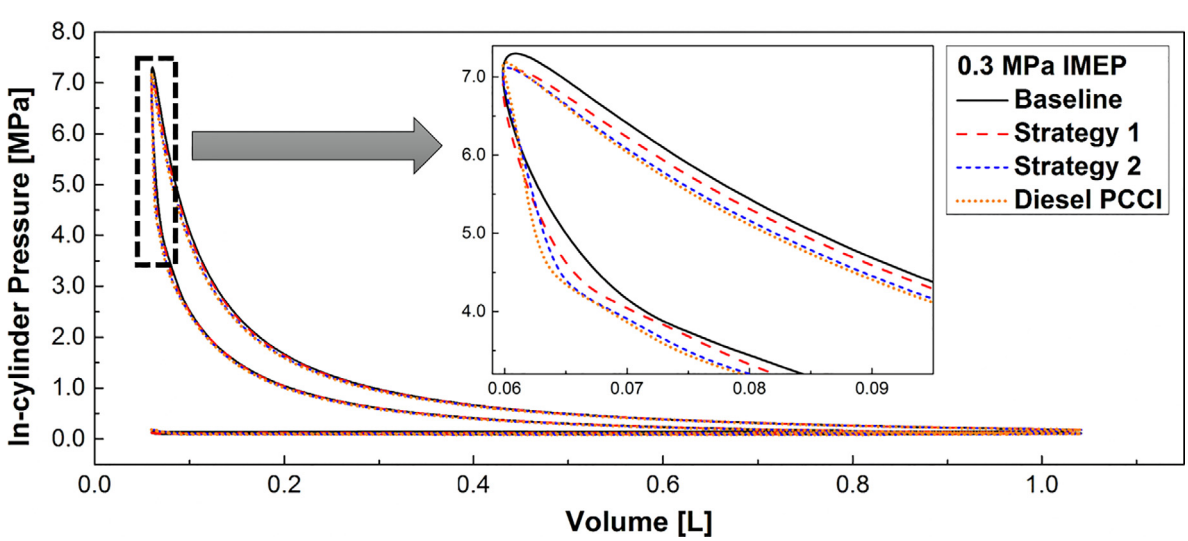


**Fig. 12.** Pressure-volume curves according to mixture stratification at 0.3 MPa IMEP.

**Table 9**
Engine operating conditions for DF-PCCI with mixture stratification.

| net IMEP [MPa] | 0.3 | 0.45 | 0.6 |
|---|---|---|---|
| NG flow rate [kg/h] | 0.16 | 0.54 | 0.82 |
| Diesel flow rate [kg/h] | 0.57 | 0.43 | 0.41 |
| Air flow rate [kg/h] | 20.3 | 27.9 | 27.7 |
| Diesel SOI [CAD aTDC] | −25 | −35 | −40 |
| NG substitution [%] | 23 | 58 | 69 |
| EGR rate [%] | 50 | 30 | 30 |
| Equivalence ratio [-] | 0.54 | 0.55 | 0.71 |

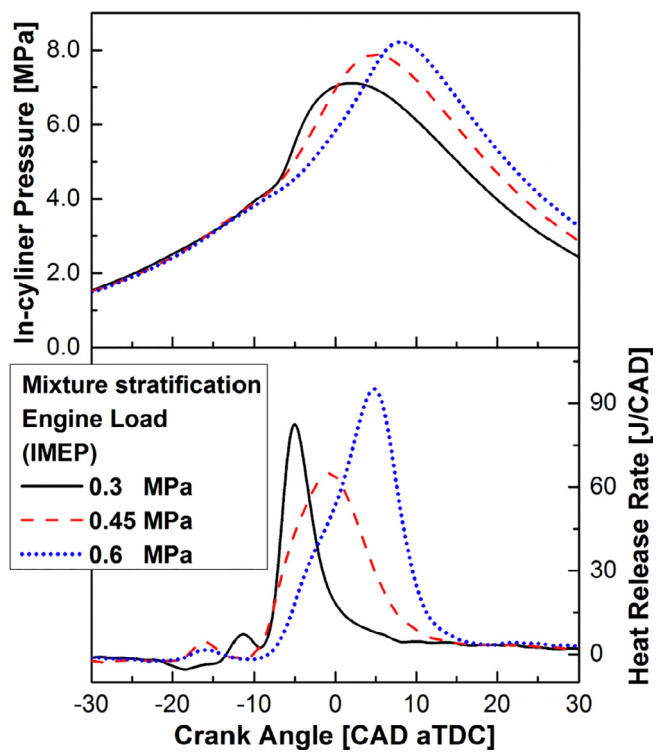


**Fig. 14.** In-cylinder pressure and heat release rate for DF-PCCI with mixture stratification over a low-load range from 0.3 to 0.6 MPa IMEP.

the heat transfer losses at 0.45 MPa IMEP.

Fig. 18 shows the THC and CO emissions for the DF-PCCI with mixture stratification and diesel PCCI at 0.3 and 0.45 MPa IMEP. The diesel PCCI effectively reduced the THC and CO emissions compared to the DF-PCCI at 0.3 MPa IMEP, because of the enhanced combustion and the absence of NG, as already explained above. However, the CO emissions were higher in the diesel PCCI than in the DF-PCCI at 0.45 MPa because of the excessive EGR rate [55].

## 4. Summary and conclusions

In dual-fuel premixed charge compression ignition (DF-PCCI) engines, the deterioration in thermal efficiency, and hydrocarbon (HC) and carbon monoxide (CO) emissions at low loads are known as the barriers for expanding the low-load operating range. In this study, the causes of the barriers were investigated and a mixture preparation strategy for overcoming the barriers was proposed and evaluated for the effectiveness in expanding the low-load limit. The experimental study was conducted in a natural gas (NG)-diesel DF-PCCI engine at a low-load range from 0.3 to 0.6 MPa indicated mean effective pressure (IMEP).

Baseline DF-PCCI was determined by combinations of diesel start of injection (SOI) and NG substitution ratio (SR). An increase in the homogeneity of a fuel–air mixture effectively reduced the $NO_X$ and PM

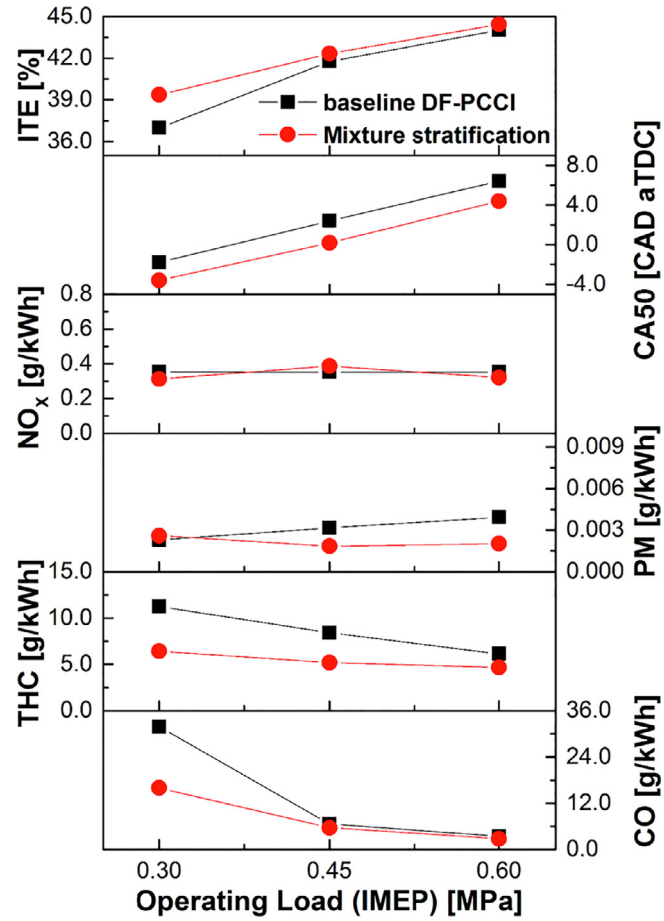


**Fig. 15.** Performance and emissions for baseline DF-PCCI and DF-PCCI with mixture stratification over a load range from 0.3 to 0.6 MPa IMEP.

emissions but increased the HC and CO emissions at each low load condition. As the engine load was decreased, the formation of an overly-lean mixture increased the mixture homogeneity further and thus aggravated the HC and CO emissions substantially. The indicated thermal efficiency (ITE) also deteriorated because the overly-lean mixture increased the combustion loss.

A mixture stratification strategy was established to increase the local equivalence ratio and reactivity of the fuel-air mixture at the low loads. The mixture stratification was realized by a retarded diesel SOI, a lower NG SR, and a higher EGR rate. The mixture stratification allowed the combustion to approach the theoretical Otto-cycle by enhancing the combustion performance, and thus the ITE was improved. The HC and CO emissions also effectively decreased mainly due to the enhanced combustion and the reduced mass flow rates of NG. The DF-PCCI with the mixture stratification improved the ITE by 6.5% and reduced the HC and CO emissions by 43% and 50%, respectively, at 0.3 MPa IMEP compared to the baseline DF-PCCI.

Diesel PCCI was also implemented at the low loads in an attempt to improve the ITE, and HC and CO emissions. Although the diesel PCCI was effective in improving the ITE at 0.3 MPa IMEP, the ITE became worse than that of the DF-PCCI at 0.45 MPa IMEP. The diesel PCCI had lower HC and CO emissions compared to the DF-PCCI at 0.3 MPa IMEP. However, the CO emissions were higher than that of the DF-PCCI at 0.45 MPa IMEP.

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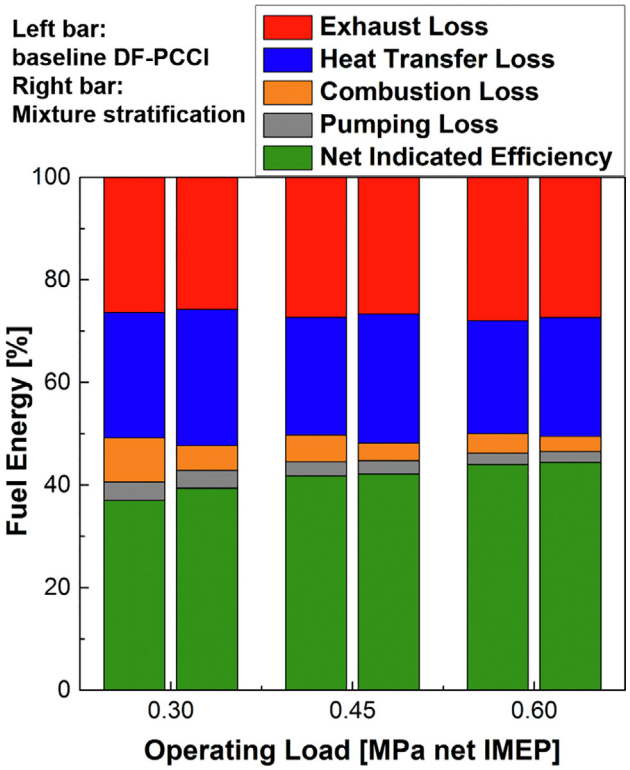


**Fig. 16.** Energy balance for baseline DF-PCCI and DF-PCCI with mixture stratification over a load range from 0.3 to 0.6 MPa IMEP.

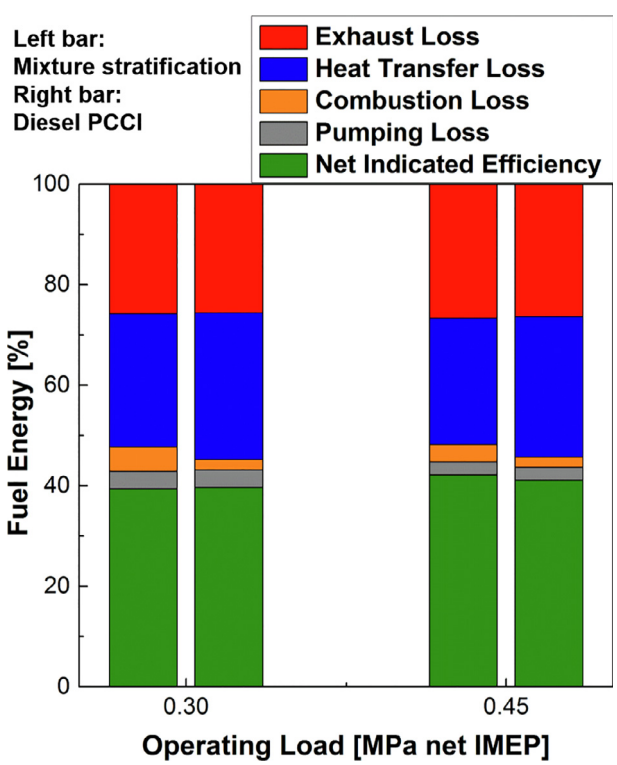


**Fig. 17.** Energy balance for DF-PCCI with mixture stratification and diesel PCCI at 0.3 and 0.45 MPa IMEP (diesel PCCI at 0.45 MPa: diesel SOI of −25 CAD aTDC, EGR rate of 65%).

Emission Vehicle (2016002070001)) of the Ministry of Environment (MOE) of Korea for financial support by the Center for Environmentally Friendly Vehicle (CEFV). The authors would also like to thank Zenobalti Co. for their technical support.

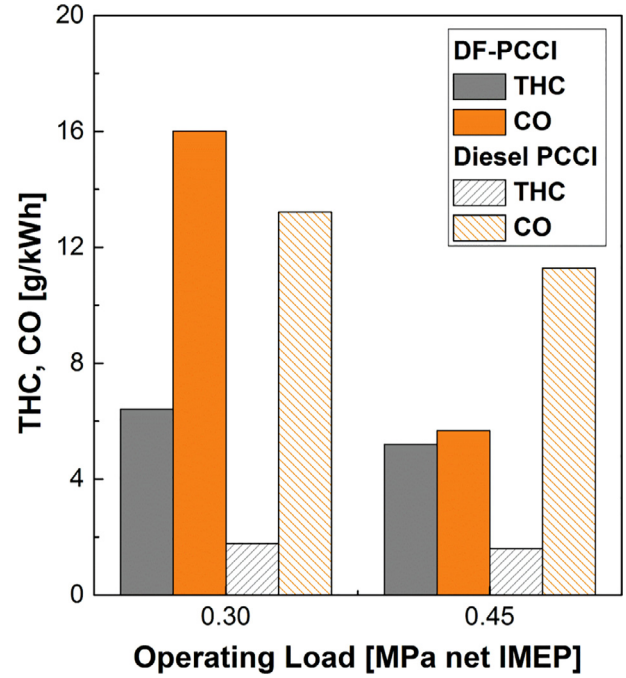


**Fig. 18.** THC and CO emissions for DF-PCCI with mixture stratification and diesel PCCI at 0.3 and 0.45 MPa IMEP (diesel PCCI at 0.45 MPa: diesel SOI of −25 CAD aTDC, EGR rate of 65%).

## References

[1] Yao M, Zheng Z, Liu H. Progress and recent trends in homogeneous charge compression ignition (HCCI) engines. Prog. Energy Combust. Sci. 2009;35(5):398–437. https://doi.org/10.1016/j.pecs.2009.05.001.
[2] Zhao H. HCCI and CAI engines for the automotive industry. Elsevier; 2007.
[3] Gan S, Ng HK, Pang KM. Homogeneous charge compression ignition (HCCI) combustion: implementation and effects on pollutants in direct injection diesel engines. Appl. Energy 2011;88(3):559–67. https://doi.org/10.1016/j.apenergy.2010.09.005.
[4] Agarwal AK, Singh AP, Maurya RK. Evolution, challenges and path forward for low temperature combustion engines. Prog. Energy Combust. Sci. 2017;61:1–56. https://doi.org/10.1016/j.pecs.2017.02.001.
[5] Bendu H, Murugan S. Homogeneous charge compression ignition (HCCI) combustion: Mixture preparation and control strategies in diesel engines. Renew. Sustain. Energy Rev. 2014 Oct;1(38):732–46. https://doi.org/10.1016/j.rser.2014.07.019.
[6] Saxena S, Bedoya ID. Fundamental phenomena affecting low temperature combustion and HCCI engines, high load limits and strategies for extending these limits. Prog. Energy Combust. Sci. 2013;39(5):457–88. https://doi.org/10.1016/j.pecs.2013.05.002.
[7] Manente V, Johansson B, Cannella W. Gasoline partially premixed combustion, the future of internal combustion engines? Int. J. Engine Res. 2011;12(3):194–208. https://doi.org/10.1177/1468087411402441.
[8] Manente V, Johansson B, Tunestal P. Partially premixed combustion at high load using gasoline and ethanol, a comparison with diesel. SAE Tech. Pap. 2009–01-0944. https://doi.org/10.4271/2009-01-0944.
[9] Lundgren MO, Wang Z, Matamis A, Andersson O, Richter M, Tuner M, et al. Effects of Post-Injections Strategies on UHC and CO at Gasoline PPC Conditions in a Heavy-Duty Optical Engine. SAE Tech. Pap. 2017–01-0753. https://doi.org/10.4271/2017-01-0753.
[10] Benajes J, Molina S, García A, Monsalve-Serrano J, Durrett R. Performance and engine-out emissions evaluation of the double injection strategy applied to the gasoline partially premixed compression ignition spark assisted combustion concept. Appl. Energy 2014;134:90–101. https://doi.org/10.1016/j.apenergy.2014.08.008.
[11] Borgqvist P, Tunestal P, Johansson B. Comparison of negative valve overlap (NVO) and rebreathing valve strategies on a gasoline PPC engine at low load and idle operating conditions. SAE Int. J. Engines (2013–01-0902) 2013;6(1):366–78. https://doi.org/10.4271/2013-01-0902.
[12] Bessonette PW, Schleyer CH, Duffy KP, Hardy WL, Liechty MP. Effects of fuel property changes on heavy-duty HCCI combustion. SAE Tech. Pap. 2007–01-0191. https://doi.org/10.4271/2007-01-0191.
[13] Reitz RD, Duraisamy G. Review of high efficiency and clean reactivity controlled compression ignition (RCCI) combustion in internal combustion engines. Prog. Energy Combust. Sci. 2015;46:12–71. https://doi.org/10.1016/j.pecs.2014.05.003.

[14] Benajes J, Molina S, García A, Monsalve-Serrano J. Effects of direct injection timing and blending ratio on RCCI combustion with different low reactivity fuels. Energy Convers. Manage. 2015;99:193–209. https://doi.org/10.1016/j.enconman.2015.04.046.
[15] Paykani A, Kakaee AH, Rahnama P, Reitz RD. Progress and recent trends in reactivity-controlled compression ignition engines. Int. J. Engine Res. 2016;17(5):481–524. https://doi.org/10.1177/1468087415593013.
[16] Inagaki K, Fuyuto T, Nishikawa K, Nakakita K, Sakata I. Dual-fuel PCI combustion controlled by in-cylinder stratification of ignitability. SAE Tech. Pap. 2006–01-0028. https://doi.org/10.4271/2006-01-0028.
[17] Tang Q, Liu H, Li M, Geng C, Yao M. Multiple optical diagnostics on effect of fuel stratification degree on reactivity controlled compression ignition. Fuel 2017;202:688–98. https://doi.org/10.1016/j.fuel.2017.04.136.
[18] Kavuri C, Paz J, Kokjohn SL. A comparison of Reactivity Controlled Compression Ignition (RCCI) and Gasoline Compression Ignition (GCI) strategies at high load, low speed conditions. Energy Convers. Manage. 2016;127:324–41. https://doi.org/10.1016/j.enconman.2016.09.026.
[19] Kokjohn SL, Hanson RM, Splitter DA, Reitz RD. Fuel reactivity controlled compression ignition (RCCI): a pathway to controlled high-efficiency clean combustion. Int. J. Engine Res. 2011;12(3):209–26. https://doi.org/10.1177/1468087411401548.
[20] Benajes J, Pastor JV, García A, Monsalve-Serrano J. The potential of RCCI concept to meet EURO VI NOx limitation and ultra-low soot emissions in a heavy-duty engine over the whole engine map. Fuel 2015;159:952–61. https://doi.org/10.1016/j.fuel.2015.07.064.
[21] Wang Y, Zhu Z, Yao M, Li T, Zhang W, Zheng Z. An investigation into the RCCI engine operation under low load and its achievable operational range at different engine speeds. Energy Convers. Manage. 2016;124:399–413. https://doi.org/10.1016/j.enconman.2016.07.026.
[22] Molina S, García A, Pastor JM, Belarte E, Balloul I. Operating range extension of RCCI combustion concept from low to full load in a heavy-duty engine. Appl. Energy 2015;143:211–27. https://doi.org/10.1016/j.apenergy.2015.01.035.
[23] Benajes J, García A, Monsalve-Serrano J, Boronat V. Achieving clean and efficient engine operation up to full load by combining optimized RCCI and dual-fuel diesel-gasoline combustion strategies. Energy Convers. Manage. 2017;136:142–51. https://doi.org/10.1016/j.enconman.2017.01.010.
[24] García A, Monsalve-Serrano J, Roso VR, Martins ME. Evaluating the emissions and performance of two dual-mode RCCI combustion strategies under the World Harmonized Vehicle Cycle (WHVC). Energy Convers. Manage. 2017;149:263–74. https://doi.org/10.1016/j.enconman.2017.07.034.
[25] Wang Y, Yao M, Li T, Zhang W, Zheng Z. A parametric study for enabling reactivity controlled compression ignition (RCCI) operation in diesel engines at various engine loads. Appl. Energy 2016;175:389–402. https://doi.org/10.1016/j.apenergy.2016.04.095.
[26] Kalsi SS, Subramanian KA. Experimental investigations of effects of EGR on performance and emissions characteristics of CNG fueled reactivity controlled compression ignition (RCCI) engine. Energy Convers. Manage. 2016;130:91–105. https://doi.org/10.1016/j.enconman.2016.10.044.
[27] Walker NR, Wissink ML, DelVescovo DA, Reitz RD. Natural gas for high load dual-fuel reactivity controlled compression ignition in heavy-duty engines. J. Energy Res. Technol. 2015;137(4):042202https://doi.org/10.1115/icef2014-5620.
[28] Nieman DE, Dempsey AB, Reitz RD. Heavy-duty RCCI operation using natural gas and diesel. SAE Int J Engines (2012–01-0379) 2012(5):270–85. https://doi.org/10.4271/2012-01-0379.
[29] Doosje E, Willems F, Baert R. Experimental demonstration of RCCI in heavy-duty engines using diesel and natural gas. SAE Tech. Pap. 2014–01-1318. https://doi.org/10.4271/2014-01-1318.
[30] Ansari E, Shahbakhti M, Naber J. Optimization of performance and operational cost for a dual mode diesel-natural gas RCCI and diesel combustion engine. Appl. Energy 2018;231:549–61. https://doi.org/10.1016/j.apenergy.2018.09.040.
[31] Hutter R, Ritzmann J, Elbert P, Onder C. Low-load limit in a diesel-ignited gas engine. Energies 2017;10(10):1450. https://doi.org/10.3390/en10101450.
[32] Li W, Liu Z, Wang Z. Experimental and theoretical analysis of the combustion process at low loads of a diesel natural gas dual-fuel engine. Energy 2016;94:728–41. https://doi.org/10.1016/j.energy.2015.11.052.
[33] Gharehghani A, Hosseini R, Mirsalim M, Jazayeri SA, Yusaf T. An experimental study on reactivity controlled compression ignition engine fueled with biodiesel/natural gas. Energy 2015;89:558–67. https://doi.org/10.1016/j.energy.2015.06.014.
[34] Park H, Shim E, Bae C. Improvement of combustion and emissions with exhaust gas recirculation in a natural gas-diesel dual-fuel premixed charge compression ignition engine at low load operations. Fuel 2019;235:763–74. https://doi.org/10.1016/j.fuel.2018.08.045.
[35] Zheng J, Wang J, Zhao Z, Wang D, Huang Z. Effect of equivalence ratio on combustion and emissions of a dual-fuel natural gas engine ignited with diesel. Appl. Therm. Eng. 2019;146:738–51. https://doi.org/10.1016/j.applthermaleng.2018.10.045.
[36] Heywood JB. Internal combustion engine fundamentals. New York: Mcgraw-hill; 1988.
[37] Stone R. Introduction to internal combustion engines. London: Macmillan; 1999. p. 543–4.
[38] Rakopoulos DC, Rakopoulos CD, Kyritsis DC. Butanol or DEE blends with either straight vegetable oil or biodiesel excluding fossil fuel: comparative effects on diesel engine combustion attributes, cyclic variability and regulated emissions trade-off. Energy 2016;115:314–25. https://doi.org/10.1016/j.energy.2016.09.022.
[39] Park H, Kim J, Bae C. Effects of Hydrogen Ratio and EGR on Combustion and Emissions in a Hydrogen/Diesel Dual-Fuel PCCI Engine. SAE Tech. Pap. 2015–01-1815. https://doi.org/10.4271/2015-01-1815.
[40] Shudo T, Nabetani S, Nakajima Y. Analysis of the degree of constant volume and cooling loss in a spark ignition engine fuelled with hydrogen. Int. J. Engine Res. 2001;2(1):81–92. https://doi.org/10.1243/1468087011545361.
[41] Benajes J, Molina S, García A, Monsalve-Serrano J. Effects of low reactivity fuel characteristics and blending ratio on low load RCCI (reactivity controlled compression ignition) performance and emissions in a heavy-duty diesel engine. Energy 2015;90:1261–71. https://doi.org/10.1016/j.energy.2015.06.088.
[42] Manente V, Johansson B, Tunestal P, Cannella W. Effects of different type of gasoline fuels on heavy duty partially premixed combustion. SAE Int J Engines (2009–01-2668) 2010;2(2):71–88. https://doi.org/10.4271/2009-01-2668.
[43] E.W. Lemmon, M.L. Huber, M.O. McLinden, NIST reference fluid thermodynamic and transport properties–REFPROP, 2002.
[44] Benajes J, García A, Monsalve-Serrano J, Villalta D. Exploring the limits of the reactivity controlled compression ignition combustion concept in a light-duty diesel engine and the influence of the direct-injected fuel properties. Energy Convers. Manage. 2018;157:277–87. https://doi.org/10.1016/j.enconman.2017.12.028.
[45] Bhaduri S, Contino F, Jeanmart H, Breuer E. The effects of biomass syngas composition, moisture, tar loading and operating conditions on the combustion of a tar-tolerant HCCI (Homogeneous Charge Compression Ignition) engine. Energy 2015;87:289–302. https://doi.org/10.1016/j.energy.2015.04.076.
[46] Rakopoulos CD, Rakopoulos DC, Kosmadakis GM, Papagiannakis RG. Experimental comparative assessment of butanol or ethanol diesel-fuel extenders impact on combustion features, cyclic irregularity, and regulated emissions balance in heavy-duty diesel engine. Energy 2019;174:1145–57. https://doi.org/10.1016/j.energy.2019.03.063.
[47] Roberts G, Rousselle CM, Musculus M, Wissink M, Curran S, Eagle E. RCCI combustion regime transitions in a single-cylinder optical engine and a multi-cylinder metal engine. SAE Int J Engines (2017–24-0088) 2017;10(5):2392–413. https://doi.org/10.4271/2017-24-0088.
[48] Papagiannakis RG, Krishnan SR, Rakopoulos DC, Srinivasan KK, Rakopoulos CD. A combined experimental and theoretical study of diesel fuel injection timing and gaseous fuel/diesel mass ratio effects on the performance and emissions of natural gas-diesel HDDI engine operating at various loads. Fuel 2017;202:675–87. https://doi.org/10.1016/j.fuel.2017.05.012.
[49] Liu J, Yang F, Wang H, Ouyang M, Hao S. Effects of pilot fuel quantity on the emissions characteristics of a CNG/diesel dual fuel engine with optimized pilot injection timing. Appl. Energy 2013;110:201–6. https://doi.org/10.1016/j.apenergy.2013.03.024.
[50] Liu J, Zhang X, Wang T, Zhang J, Wang H. Experimental and numerical study of the pollution formation in a diesel/CNG dual fuel engine. Fuel 2015;159:418–29. https://doi.org/10.1016/j.fuel.2015.07.003.
[51] Wei L, Geng P. A review on natural gas/diesel dual fuel combustion, emissions and performance. Fuel Process. Technol. 2016;142:264–78. https://doi.org/10.1016/j.fuproc.2015.09.018.
[52] Divekar PS, Chen X, Tjong J, Zheng M. Energy efficiency impact of EGR on organizing clean combustion in diesel engines. Energy Convers. Manage. 2016;112:369–81. https://doi.org/10.1016/j.enconman.2016.01.042.
[53] Uchida N, Okamoto T, Watanabe H. A new concept of actively controlled rate of diesel combustion for improving brake thermal efficiency of diesel engines: Part 1—verification of the concept. Int. J. Engine Res. 2018;19(4):474–87. https://doi.org/10.1177/1468087417720332.
[54] Rakopoulos CD, Rakopoulos DC, Mavropoulos GC, Kosmadakis GM. Investigating the EGR rate and temperature impact on diesel engine combustion and emissions under various injection timings and loads by comprehensive two-zone modeling. Energy 2018;157:990–1014. https://doi.org/10.1016/j.energy.2018.05.178.
[55] Chen Z, Wu Z, Liu J, Lee C. Combustion and emissions characteristics of high n-butanol/diesel ratio blend in a heavy-duty diesel engine and EGR impact. Energy Convers. Manage. 2014;78:787–95. https://doi.org/10.1016/j.enconman.2013.11.037.